1. The purpose of this memo

Since I did a lot of the explaining the story and the idea behind the dashboard in the presentation and tailoring it to the audience and making it tired to an audience that is less experienced in the crypto and finance world, I wanted to make the memo a bit more about the methodological choices I made, how I collected the data, built off of my past projects all of that so that is what I aim to do through this written portion of the project. I want to keep it brief and just supplement the work I did on the website and the final presentation with this memo.

2. How the Pattern matching works

Unlike the project I built off of, this is based on two data sources: Yahoo Finance daily candles back to 2014, and a Bitstamp hourly one-time CSV that I downloaded with hourly candles back to 2018. I use Yahoo Finance to get a little bit of edge on each refresh. The dashboard refreshes itself on a GitHub Actions cron every 12 hours, so the data stays current without me touching anything between refreshes.

From the past project, I brought over a few features like returns, lagged returns, rolling means, and standard deviations. Each column in each feature is rolling z-scored over a 30-day window before they're compared, and I used z-score to get past the absolute price level. So prices in 2018 and 2025 can be compared for shape, not just the numbers.

I used cosine distance as the similarity metric because it's sensitive to shape, but not so much the magnitude, which is what I needed when comparing the shapes of past patterns. Then I use the top ten nearest neighbors to become matches for that window.

The pattern matching runs at four lookback sizes: 12 hours, 24 hours, 72 hours, and one week. The audience can switch between them on the dashboard depending on the time horizon that matches their question. The shorter three windows use the hourly data; the one-week window uses the daily data so it draws from a longer history.

One thing that I came across early was that the forward returns were briefly being calculated from z-scores instead of raw prices. So it would say, like, a 200% increase in 24 hours when the actual move was -2%. Thankfully I found this early on and was able to fix it in the starting stages of the project.

There were also a couple constraints that I wanted to mention here. The hourly history only goes back to 2018, not earlier, because that is what the free Bitstamp CSV provides. Markets change regimes (halvings, ETF approval, regulation), so a 2018 match may not generalize to 2025 conditions. The dashboard shows ranges instead of point estimates precisely because based on the analysis I cannot ethically claim anything more.

3. My thinking behind the final structure

For the structure, I wanted to tell a data -driven story, so I included the simpler tableau visuals at the start to kind of get people familiar with Bitcoin data, what it looked like over time, and what it looks like daily. Then I chose to move on to the slightly more complicated pattern prediction and recognition window so that readers would know how to interpret it a little bit more.

I chose to have them in a carousel format just so it would be less scrolling and a little bit more aesthetic. And along with the color, I wanted to use the same color scheme as the Bitcoin logo or Bitcoin had in my mind, which was the dark gray, the orange, and a little bit of white. also included text throughout the website to keep it a combination of visuals and writing for the most understanding for less experienced people.

I chose to keep the language as simple as I could with any jargon that I used explained because there is a little bit of very basic jargon in cryptocurrency. I was also a little inspired by the layout of Microsoft365, the dashboard page for that. So I tried to take a little bit of inspiration from that and try to make it a little bit clean and aesthetic like it is

4. Building it and what's next in my mind

So as per this semester and the rubric for this final project, I did not use any generative AI to come up with any of the storytelling or write anything for me. But I did use Claude as a little bit of a coding assistant for more of the technical work like scaffolding the Next. js app and helping me set up the Python pipeline. I used it to wire some components together like getting the webhooks for the Tableau visuals linked together because they all use the same link. And I also wanted to try something new for the front end to make it look good. So I used the new Claude design tool that recently came out where I gave it some, uh, screenshots of my existing old dashboard and kind of described what I wanted and what I wanted to look like and things like that. And it gave me back some files that I then integrated into a branch in my repository and then combined that branch with domain main at the end.

All the decisions about the audience story, what to include, what to leave out, and all the writing stuff was all mine and done by me. I just used Claude because I have experience setting up websites and things like that using that tool outside of an academic situation. And I also picked option four to kind of be able to build on those skills and build something I can work on in the future and continue to develop.

Something I wanna do in the future with this is I want to add some other digital assets, maybe, like, Ethereum or Solana, and maybe even some stocks and things like that, not necessarily cryptocurrency. I'd also like to get a longer window pattern, um, maybe multi months or something like that as the daily history accumulates.